Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/14/2005
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Project Title:	A Novel Biodosimetry Method For Monitoring Radiation-Induced Genetic Damage		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53711-5399	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/01/2003	**End Date:**	06/01/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1525		
Performance Goal No.:			
Performance Goal Text:			
	For extended space missions, exposure to hazardous radiation from galactic cosmic radiation and solar particle events poses a serious threat to the health of the astronauts. Dosimeters measure radiation dose received but not its biological effects or individual differences in radiation sensitivity. Accurate methods for measuring the extent of an individuals genetic damage resulting from accumulated radiation exposure is critical for estimating personal health risks and need for countermeasures. The overall goal of this proposal is to investigate the efficacy of a novel biodosimetry method for monitoring mutational load resulting from radiation-induced mutations in repetitive DNA sequences. Repetitive sequences constitute nearly one third of the human genome and some have been shown to be extremely sensitive to radiation-induced mutations. Most repetitive sequences are in non-coding regions of the genome and, therefore, alterations in these loci are usually not deleterious to the survival of the host cell. Thus, mutations in non-coding repetitive sequences can accumulate, providing a stable molecular record of DNA damage from all past exposures.		

Task Description:	Progenitor stem cells are long lived and continually divide producing identical duplicate progeny cells. Mutations in repetitive DNA loci in progenitor stem cells will accumulate and through analysis of progeny blood or sperm cells can serve as biological sensors to monitor radiation-induced genetic damage. To test this hypothesis we will (a) screen a broad range of highly mutable repetitive DNA sequences to identify genetic loci most sensitive to high-LET radiation-induced mutations, (b) show that radiation-induced mutations in progenitor stem cells are cumulative and that mutational load can be quantitatively measured by sampling a population of single or small pools of progeny blood or sperm cells for changes in repetitive DNA sequences, and (c) investigate possible mechanisms causing radiation-induced genomic instability in repetitive DNA loci. Knowledge obtained from completion of this proposal will determine the efficacy of a novel quantitative method for monitoring radiation-induced mutational load that would be applicable to both mouse models and humans. Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of radiation-induced mutations that would be useful in predicting and minimizing health risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of mutations that would be useful in predicting and minimizing health risks for astronauts or any individual exposed to ionizing radiation or chemical mutagens. Preliminary data suggest that the markers identified in this study are also sensitive biomarkers for oxidative stress and may be useful in early detection of conditions that could lead to cancer and other degenerative diseases and may be useful in the helping to develop effective countermeasures.
Task Progress:	The first aim of this study is to identify DNA repeat markers in human and mouse genomes that are sensitive to radiation-induced mutations that can be used for biodosimetry. Mouse and human cell lines with and without functional DNA mismatch repair were irradiated with 0.5, 1 or 3 Gy 1 GeV iron ions at 0.5Gy/min. Total genomic DNA was extracted and quantified from cells collected at 24, 48 and 72hr post-irradiation. The DNA was diluted to give roughly 1-2 genome equivalents (6-12pg), amplified by small-pool PCR then analyzed by fluorescent capillary electrophoresis to identify mutant alleles. Markers being tested include mono-, tri-, tetra-, penta- and hexanucleotide repeats on autosomal and X/Y chromosomes and in the mitochondrial genome. Repeat markers identified that are sensitive to radiation-induced mutations include: (1) human and mouse mononucleotide repeats consisting of long polyA tracts (1-3% mutations frequency in some markers) (2) some tetranucleotide repeats on human Y chromosome (0.5 to 4% mutations frequency), (3) some pentanucleotide repeats in both mouse and humans (1-4% mutation frequency). Use of mismatch repair deficient cells resulted in unacceptably high background mutation rates despite attempts to minimize problem by starting cultures from single cells. We are now using cell lines derived from normal tissues which produce very low levels of spontaneous mutant alleles. The validity of the small-pool PCR approach was confirmed by comparing mutation rates to those observed in 200 single cell clones for same irradiated cell cultures. Our results indicate that detection of radiation-induced mutations in selected repeat sequences is sufficiently high to make it feasible to detect using small-pool PCR approach and preliminary data suggest a dose response. Human and mouse multiplex assays containing 5-6 of the most sensitive markers are being constructed to increase throughput for future experiments. Preliminary experiments for the second aim to show that radiation-induced mutations in progenitor stem cells are cumulative and that they can be measured by sampling a population of progeny blood or sperm cells for changes in repetitive DNA sequences are underway. C57Bl\6 mice were irradiated with 0.5, 1 or 3 Gy 1 GeV iron ions at 0.5Gy/min then maintained for 10 weeks post irradiation to allow turn over of sperm cells. We are currently testing for mutations in selected DNA repeat markers for mutations in blood, sperm, buccyl cells and a number of organs. Larger experiments are planned for summer of 2005 and some of those mice will receive multiple exposures over a period of several months. Mice over and under expressing antioxidant enzyme Gpx4 or the Msh2 mismatch repair gene are being crossed with mutation reporter mice containing the SupFG1 lambda transgene for irradiation experiments in the summer of 2005. These mice will allow us to investigate the role of oxidative stress and DNA repair in radiation-induced instability of repetitive DNA sequences. Preliminary experiments with wildtype C57Bl\6 mice treated with parquat to induce oxidative stress revealed that extended mononucleotide repeats that were sensitive to radiation-induced mutations were also very sensitive to ROS-induced mutations, indicating the possible mechanism for mutagenesis for these markers. We speculate that mutations in repeat DNA sequences are primarily due to indirect radiation damage resulting from generation of ROS and that therefore mutagenesis in this type of marker may be radiation-type independent.
Bibliography Type:	Description: (Last Updated: 04/16/2019)
Articles in Peer-reviewed Journals	Bacher, J.W.; Flanagan, L.A.; Smalley, R.L.; Nassif, N.A.; Burgart, L.J.; Halberg, R.B.; Megid, W.M.; Thibodeau, S.N "Development of a fluorescent multiplex assay for detection of MSI-High tumors" Disease Markers 20:237-250 , Nov-2004
Articles in Peer-reviewed Journals	Bacher, J.; Megid, W.; Kent, M.; Halberg, R "Use of Mononucleotide Repeat Markers for Detection of Microsatellite Instability in Mouse Tumors" Molecular Cancinogenesis (submitted) , Jan-2005
Patents	Application #60/621277 Provisional patent application Oct-2004 Bacher, J; Halberg, R; Kent, M "METHODS AND KITS FOR DETECTING MUTATIONS"
Presentation	Bacher, J.; Halberg, R.; Megid, W.; Kent, M.; Prolla, T. "Monitoring Radiation-Induced Genetic Damage" 3rd International Workshop on Space Radiation Workshop and 15th Annual NASA Space Radiation Health Investigators' Workshop. May-2004
Presentation	Bacher, J.; Halberg, R.; Megid, W.; Prolla, T "A Novel Biodosimetry Method for Monitoring Radiation-Induced Genetic Damage" NASA Bioastronautics Investigators Workshop, Galvenston, TX Jan-2005